

InstaSHAP Reproducibility Report

Objective: reproduce the key InstaSHAP tabular experiments from the ICLR 2025 paper.

Datasets: Bike Sharing (synergy), Covertypes (redundancy), Adult Income (supplementary).

Methodology: black-box baseline, additive GAMs, masked surrogate, and InstaSHAP Eq. (20).

Data source implementation: `ucimlrepo.fetch_ucirepo(id=275 / 31 / 2)`.

Outputs: metrics tables, plots, SHAP vs InstaSHAP comparison, and paper-reference tables.

Adult Overview

Dataset: adult

Task: classification

Features: age, fnlwgt, capital_gain, capital_loss, hours_per_week, workclass, education, marital_status, occupation, relation

Interaction pairs: []

Implementation notes:

- Black-box model: MLP by default, with optional RandomForest.
- GAM-1/GAM-2: additive neural subnetworks over feature groups.
- InstaSHAP: masked additive training against a learned masked surrogate.
- SHAP: permutation explainer aggregated back to original feature groups.

Adult Metrics

model	accuracy	log_loss	training_seconds	inference_seconds_mean
blackbox	0.8400	0.3685	0.2628	0.0006
gam1	0.8362	0.3750	0.9782	0.0041
instashap	0.8400	0.4235	1.4262	0.0044

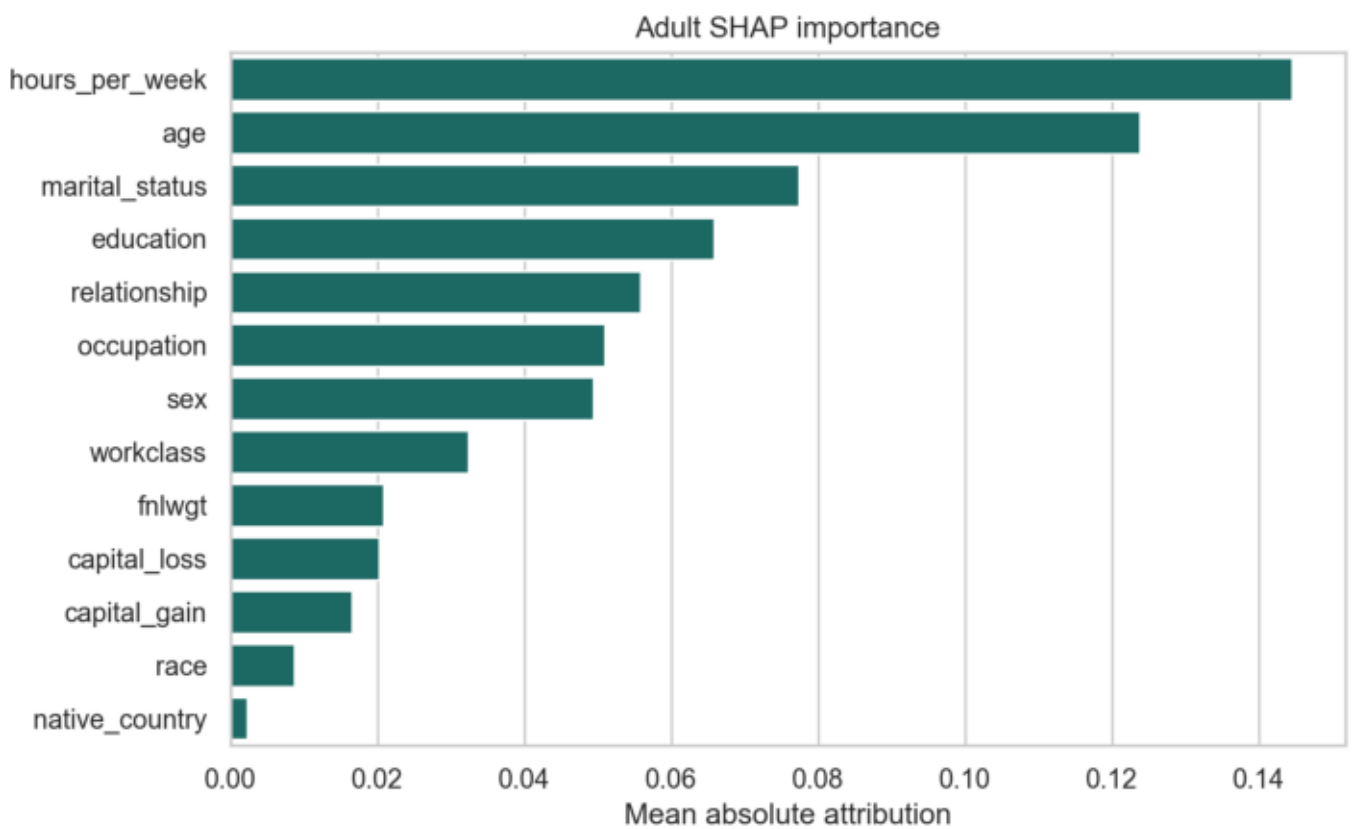
Adult vs Paper

	model	reproduced	paper
gam1	0.8362	0.842	
instashap	0.8400	0.843	

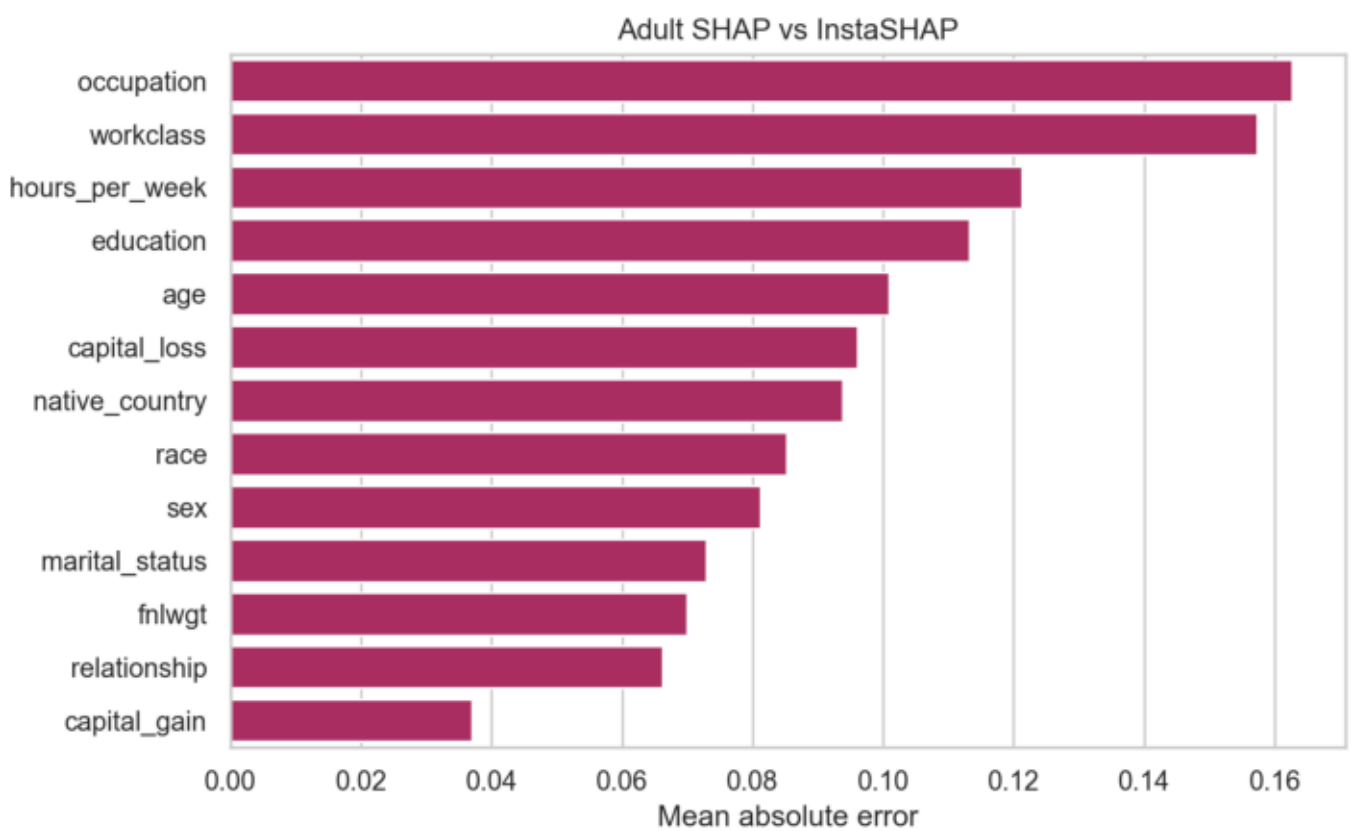
Adult Explanation Comparison

model	training_seconds	seconds_total	samples	mse	mae
surrogate	0.7629	NaN	NaN	NaN	NaN
shap	NaN	0.3945	24.0	NaN	NaN
instashap	NaN	0.0050	24.0	NaN	NaN
shap_vs_instashap	NaN	NaN	NaN	0.012	0.0968

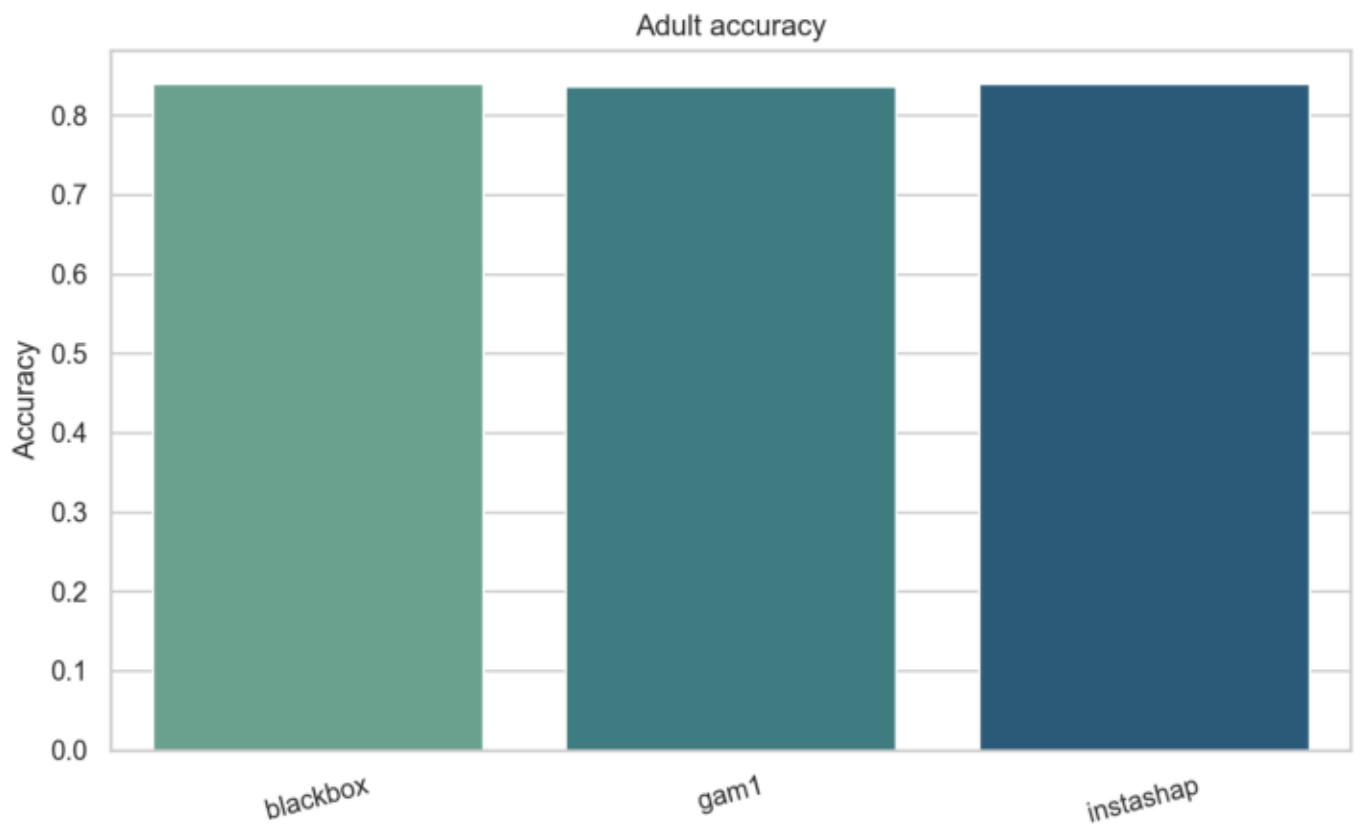
Adult Plot



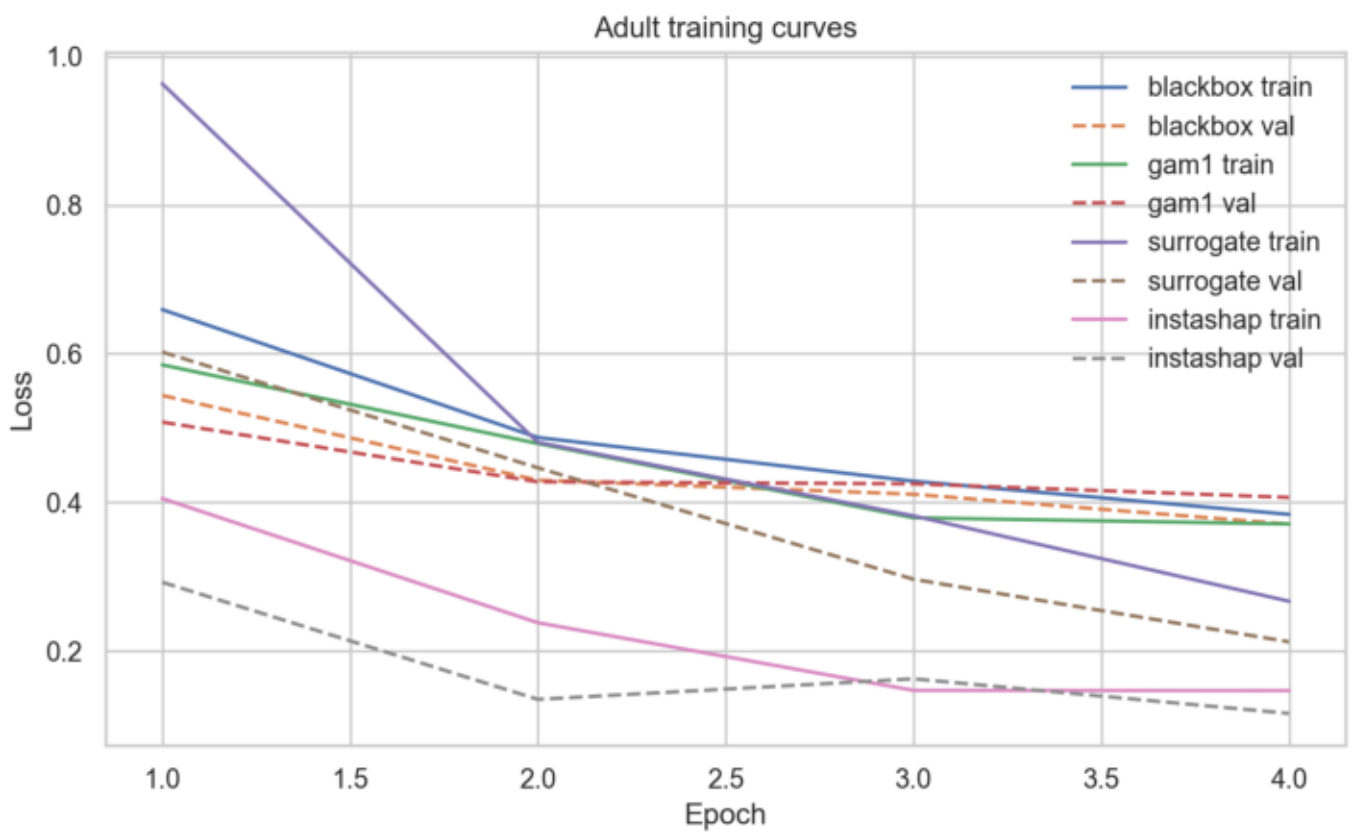
Adult Plot



Adult Plot



Adult Plot



Bike Overview

Dataset: bike

Task: regression

Features: day_of_month, temp, atemp, hum, windspeed, season, year, month, hour, holiday, weekday, workingday, weather.

Interaction pairs: [['hour', 'workingday']]

Implementation notes:

- Black-box model: MLP by default, with optional RandomForest.
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- SHAP: permutation explainer aggregated back to original feature groups.

Bike Metrics

model	rmse	mse	r2	nmse_pct	training_seconds	inference_seconds_mean
blackbox	265.3422	70406.4766	-1.0485	204.8466	3.5736	0.0015
gam1	254.7480	64896.5312	-0.8882	188.8155	1.5747	0.0061
gam2	253.8135	64421.2734	-0.8743	187.4328	1.8322	0.0063
instashap	264.7821	70109.5781	-1.0398	203.9828	2.4168	0.0066

Bike vs Paper

	model	reproduced	paper
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blackbox	204.8466	6.59	
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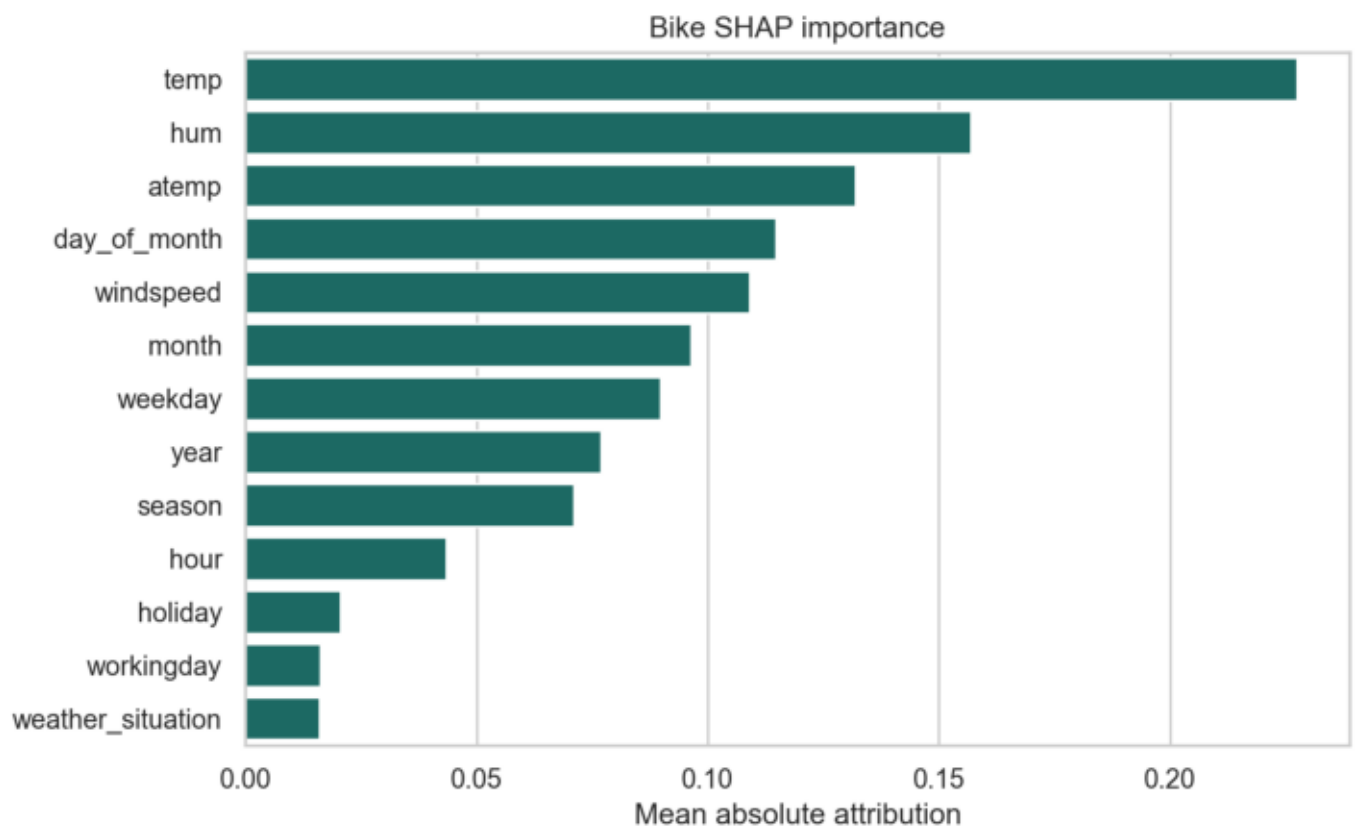
gam1	188.8155	17.40	
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gam2	187.4328	6.23	
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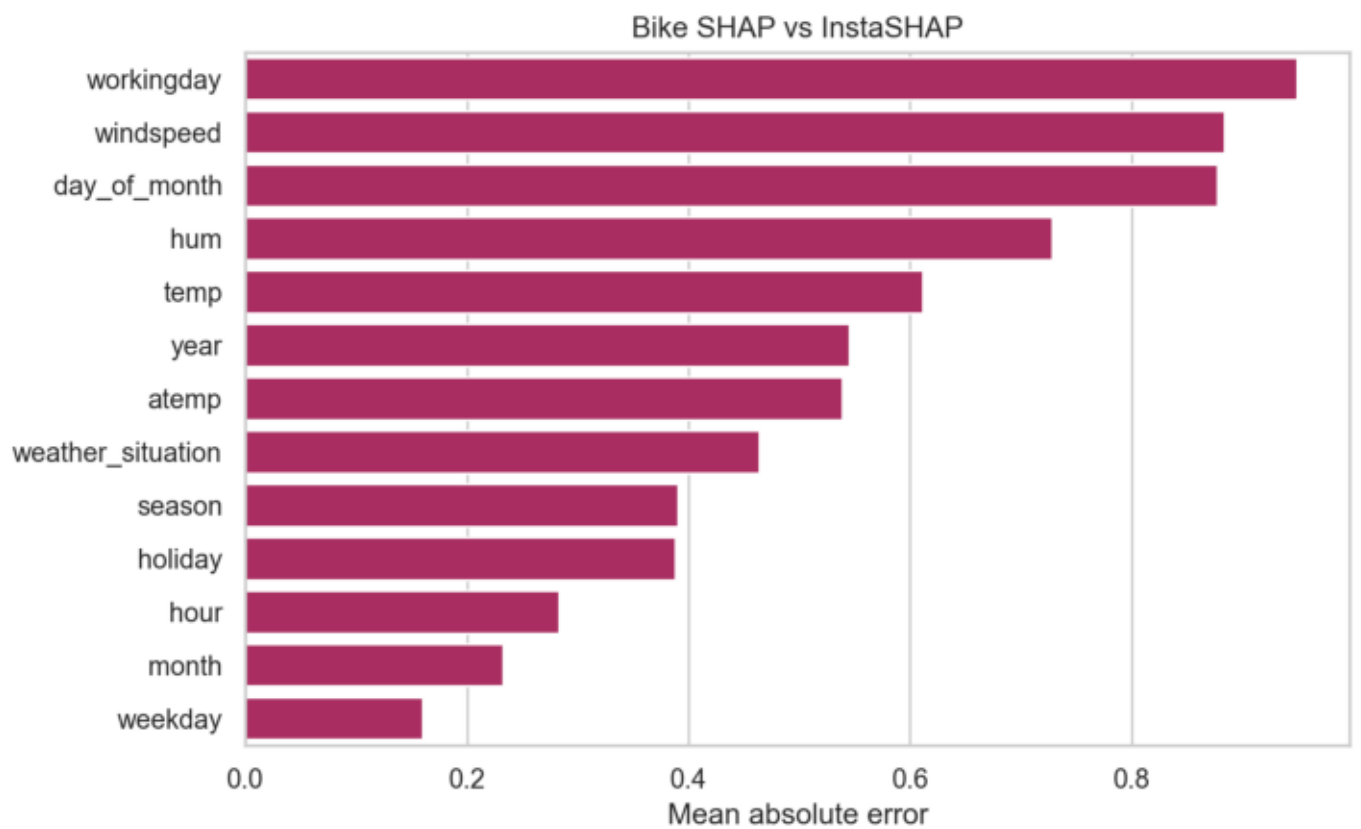
Bike Explanation Comparison

model	training_seconds	seconds_total	samples	mse	mae
surrogate	1.0681	NaN	NaN	NaN	NaN
shap	NaN	12.6959	32.0	NaN	NaN
instashap	NaN	0.0083	32.0	NaN	NaN
shap_vs_instashap	NaN	NaN	NaN	0.3712	0.5427

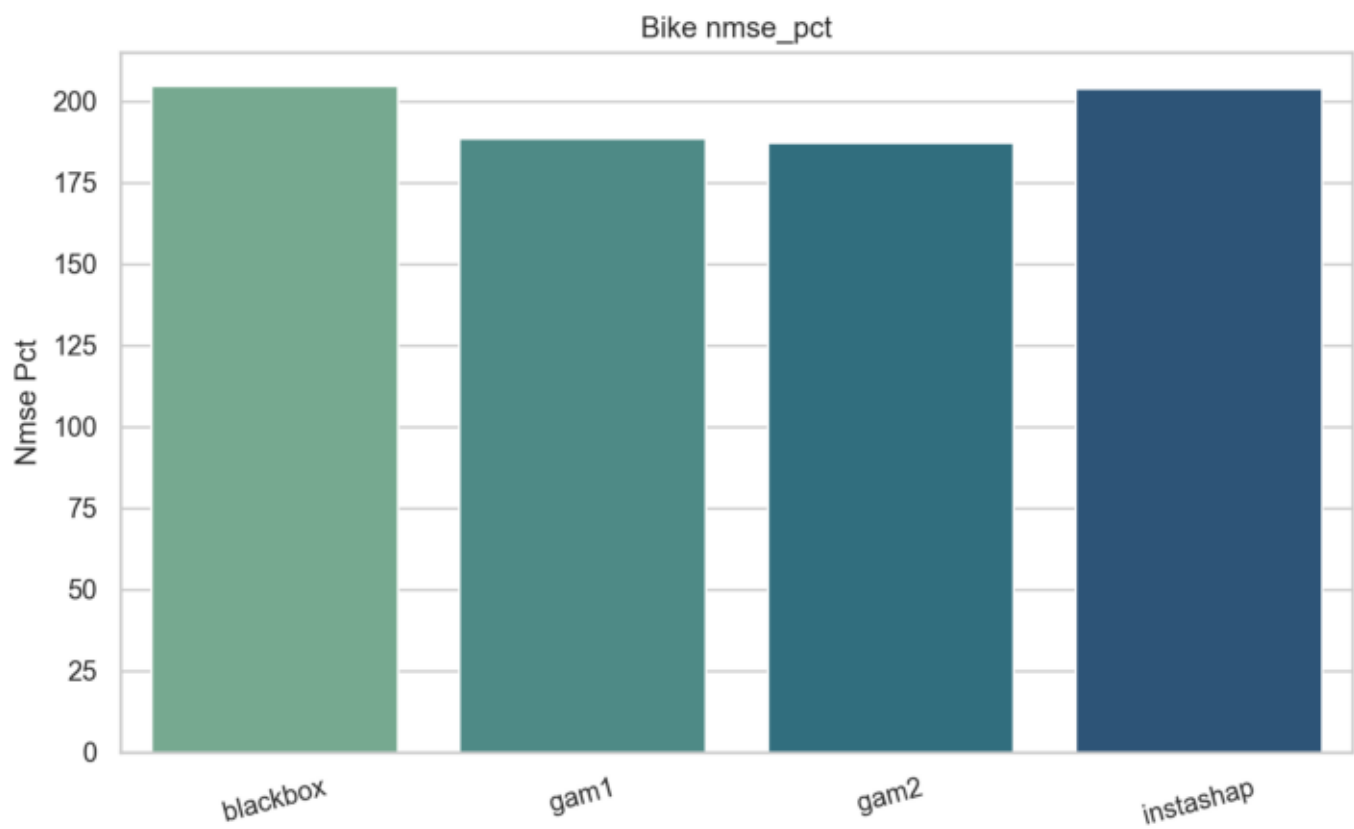
Bike Plot



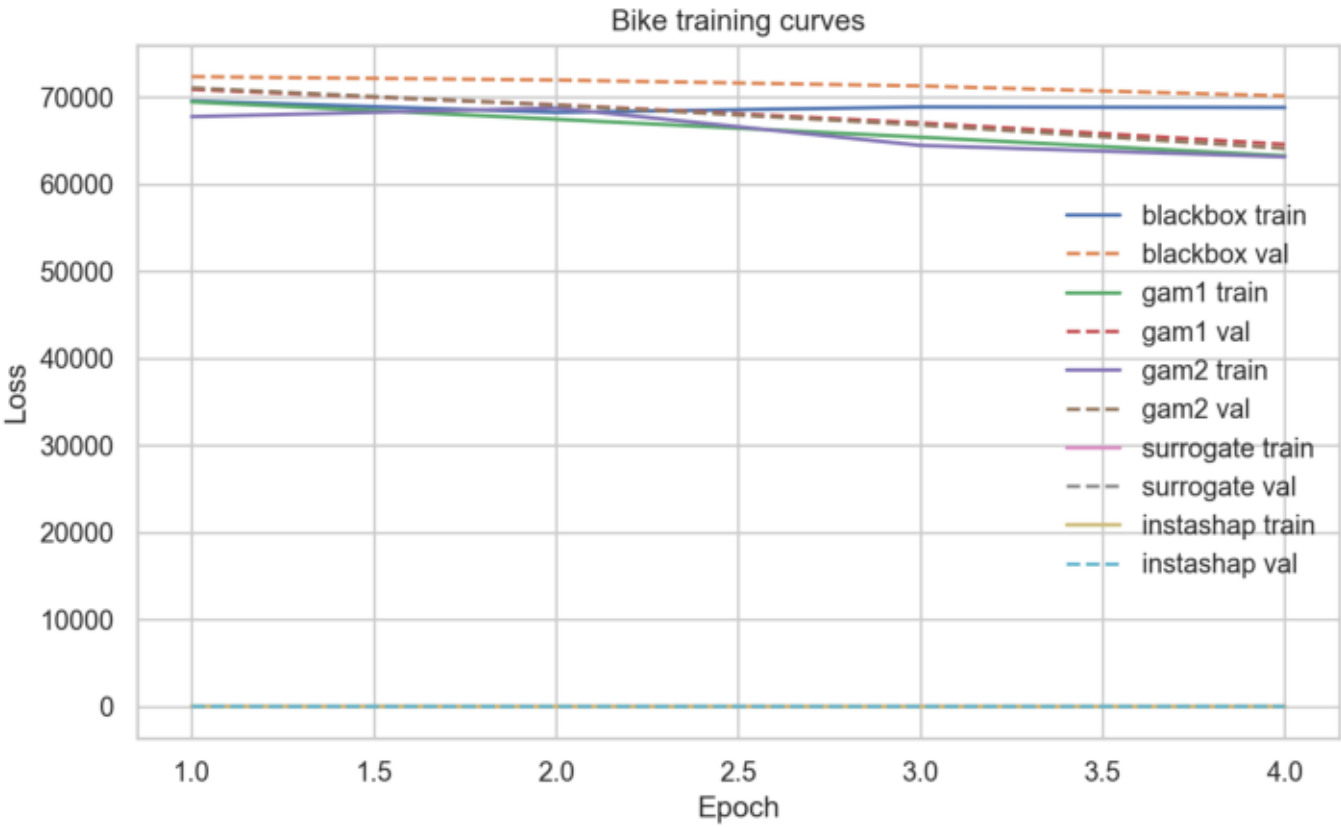
Bike Plot



Bike Plot



Bike Plot



Covertypes Overview

Dataset: covertypes

Task: classification

Features: elevation, aspect, slope, horizontal_distance_to_hydrology, vertical_distance_to_hydrology, horizontal_distance_to_hydrology

Interaction pairs: [['elevation', 'soil_climate_zone']]

Implementation notes:

- Black-box model: MLP by default, with optional RandomForest.
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- SHAP: permutation explainer aggregated back to original feature groups.

Covertypes Metrics

model	accuracy	log_loss	training_seconds	inference_seconds_mean
blackbox	0.6375	0.9931	0.8413	0.0007
gam1	0.6688	0.8895	1.3644	0.0058
gam2	0.6912	0.8041	1.5903	0.0070
instashap	0.4875	1.3345	2.0798	0.0056

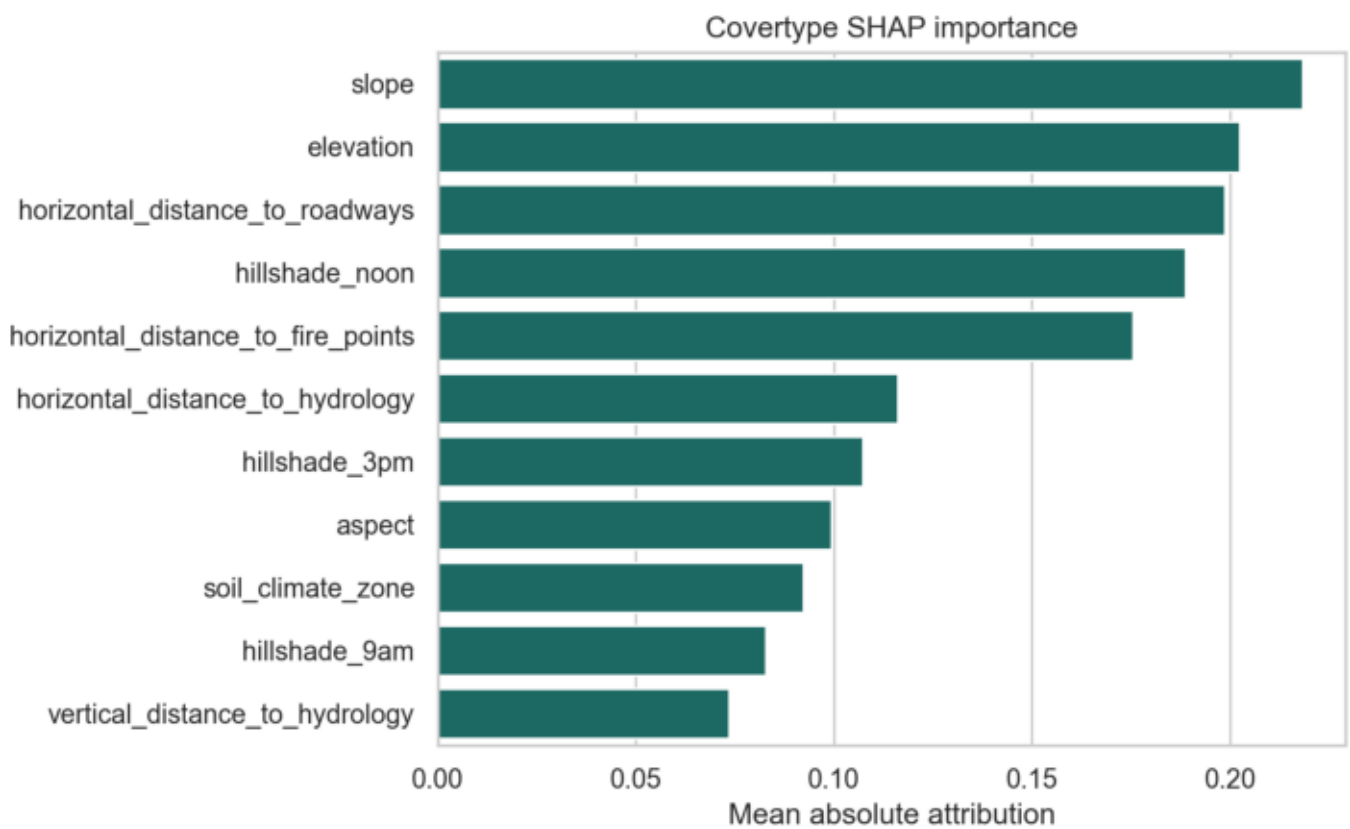
Covertypes vs Paper

	model	reproduced	paper
blackbox	0.6375	0.804	
gam1	0.6688	0.724	
gam2	0.6912	0.822	

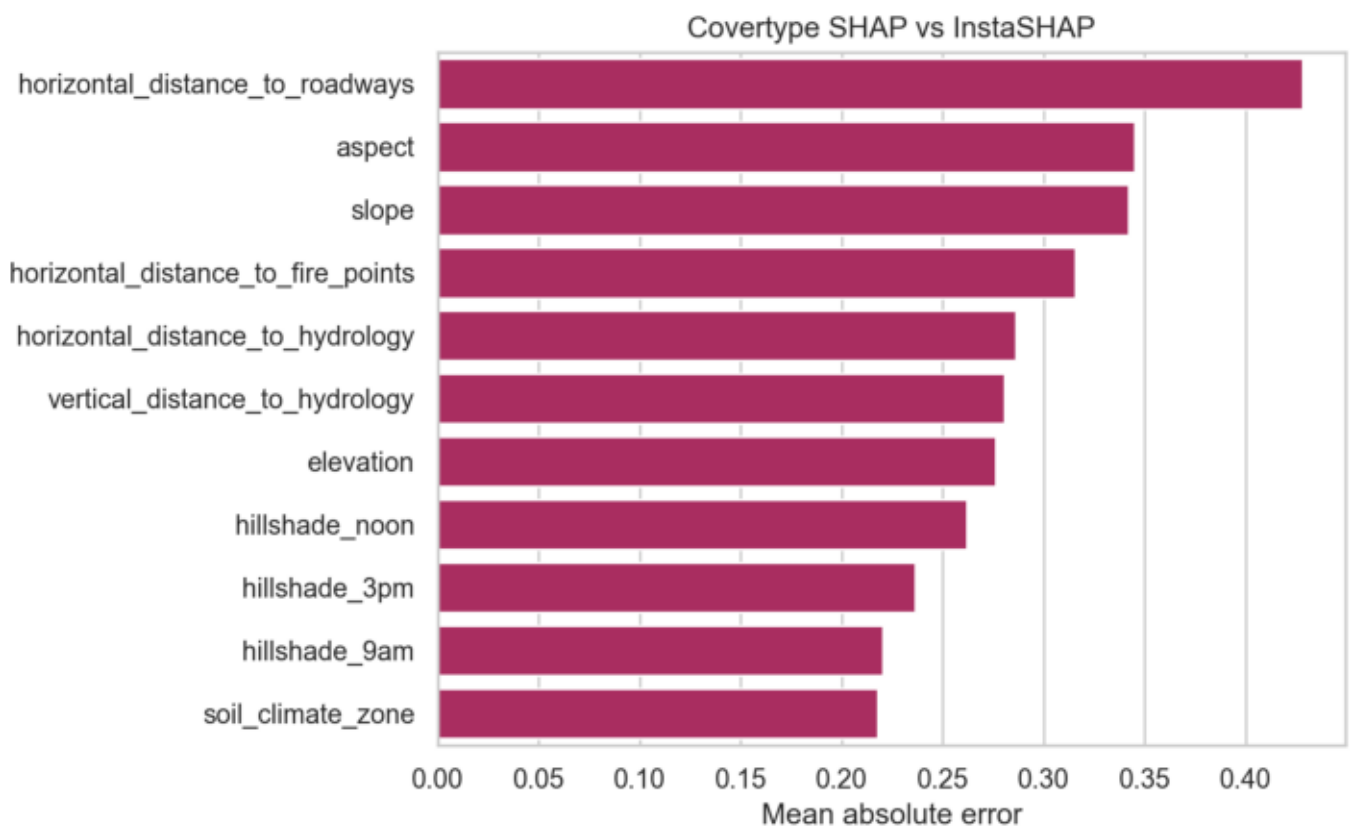
Covertypes Explanation Comparison

model	training_seconds	seconds_total	samples	mse	mae
surrogate	1.3161	NaN	NaN	NaN	NaN
shap	NaN	0.5957	24.0	NaN	NaN
instashap	NaN	0.0073	24.0	NaN	NaN
shap_vs_instashap	NaN	NaN	NaN	0.1099	0.2919

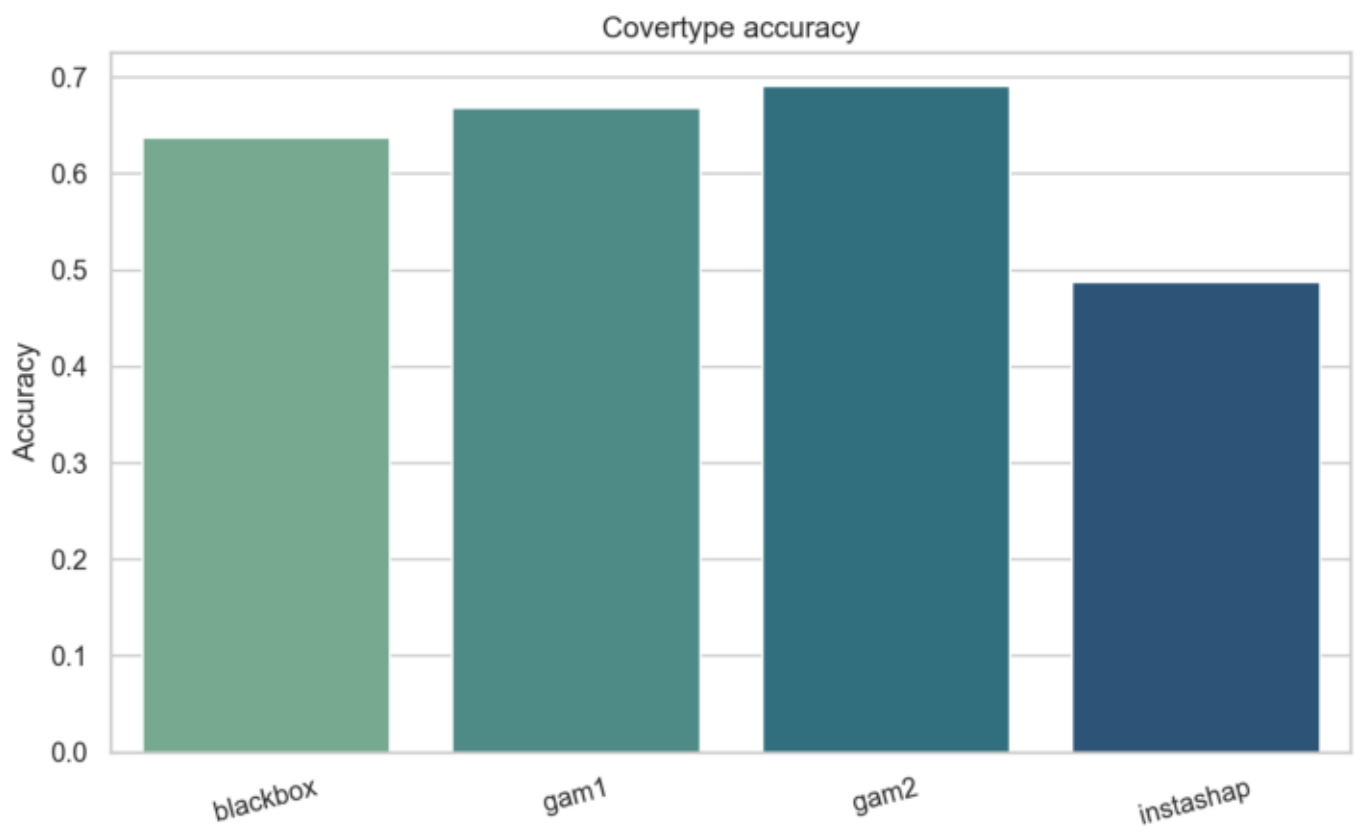
Covertime Plot



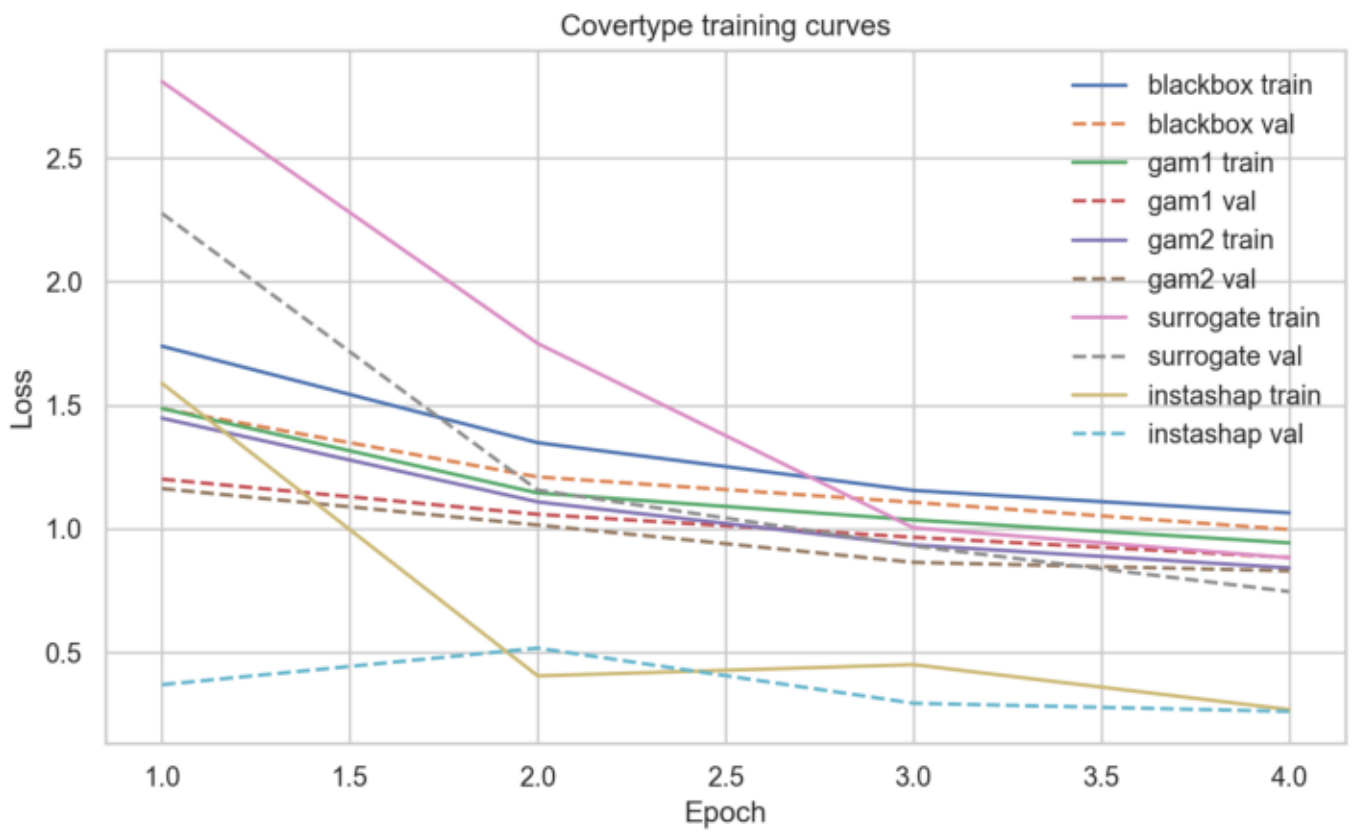
Covertypes Plot



Covertypes Plot



Covertypes Plot



Comparison and Observations

Observations:

- The report juxtaposes reproduced metrics against the values quoted in the paper.
- Remaining gaps usually come from limited epochs, smaller subsets, and missing original hyperparameters.
- InstaSHAP artifacts include both predictive performance and explanation-fidelity measurements.

Recommendation:

Increase epochs and Covertypes sample size for a closer numerical match to the paper before final benchmarking.